

Surface Effects on Playing Style in 2013 Grand Slam Tennis

An exploratory analysis of court surface, gender, and tournament round effects on professional tennis playing style

The Data

This analysis uses the 2013 Grand Slam match statistics dataset from the UCI Machine Learning Repository, which contains match-level performance statistics from the Australian Open, French Open, Wimbledon, and US Open, for both men's and women's singles. Each row is a match, with separate statistics recorded for the two players (e.g. aces, first-serve percentage, winners, unforced errors), alongside tournament, round, and outcome. Combined across all eight tournament/gender files, the dataset covers 1,010 matches.

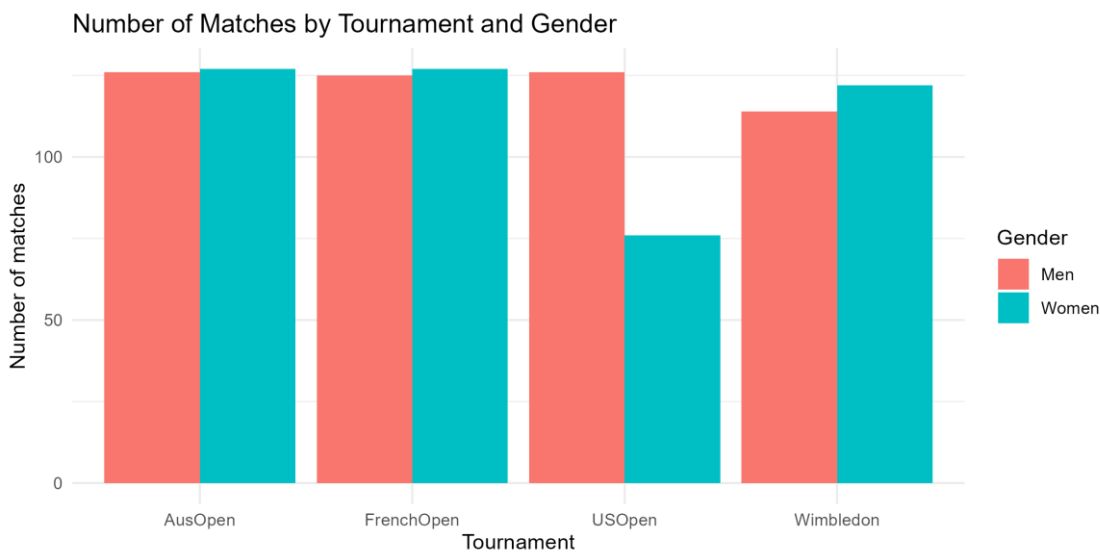


Figure 1. Number of matches per tournament and gender. Coverage is broadly even across tournaments and genders, except for the US Open women's draw, which has noticeably fewer recorded matches.

A derived **Surface** variable groups the four tournaments into Clay (French Open), Grass (Wimbledon), and Hard (Australian Open, US Open), which is the main lens used throughout this analysis.

Data dictionary

The abbreviations used throughout this report:

Abbreviation	Meaning
ACE	Aces served
FSP	First serve percentage (%)
FSW	First serve points won
TPW	Total points won
UFE	Unforced errors committed
WNR	Winners hit

A Note on Player Labels: Winner vs Loser

The raw dataset labels players as “Player 1” and “Player 2”, but this label carries no consistent meaning because P1 is not always the match winner. Comparing statistics as “P1 minus P2” therefore mixes, within the same column, some rows where the winner's advantage is being measured and other rows where the loser's disadvantage is being measured, depending on who happened to be listed first. To fix this, every paired statistic (e.g. ACE.1 / ACE.2) is recoded into winner (_W) and loser (_L) columns based on the match result, and a margin column (winner – loser) is derived from each pair. Every margin in this report therefore has one consistent meaning: how much better the winner performed than the player they beat.

Data Quality and Missingness

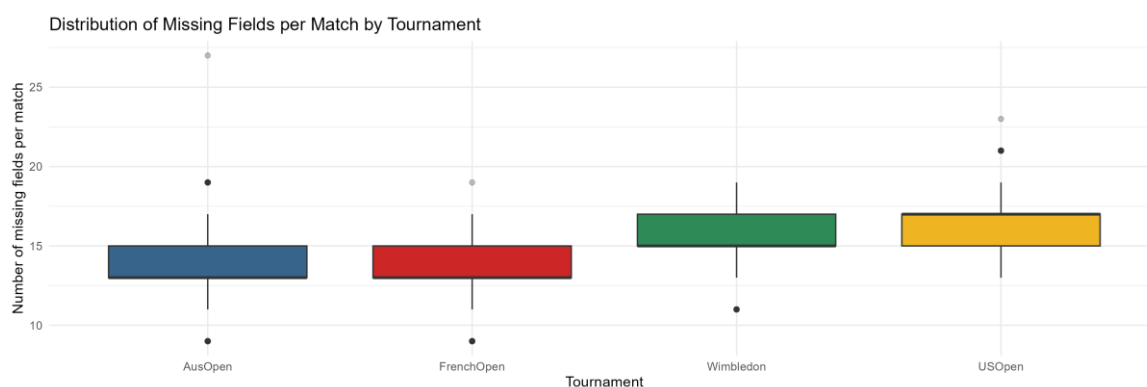


Figure 2. Distribution of missing fields per match, by tournament. Against the dataset total of 1,010 matches, the US Open and Wimbledon show noticeably higher median missingness (around 15–17 fields per match) than the Australian and French Opens (around 13–15). The US Open is also the tournament with the fewest recorded matches overall (Figure 1), and its round information in particular is much sparser than the other three majors. Both issues are treated explicitly later: all analysis below uses available-case data, and the round-by-round breakdown excludes the US Open entirely.

The dataset is otherwise machine-readable and consistently formatted, but the underlying metadata does not document why particular fields are missing for particular tournaments, which limits how confidently the missingness itself can be interpreted (e.g. whether it reflects different recording practices by tournament, or genuinely unavailable statistics).

Surface Effects on Playing Style

To look at playing style at the player level, each match contributes two observations (winner and loser), for four key statistics: ACE, FSP, UFE, and WNR.

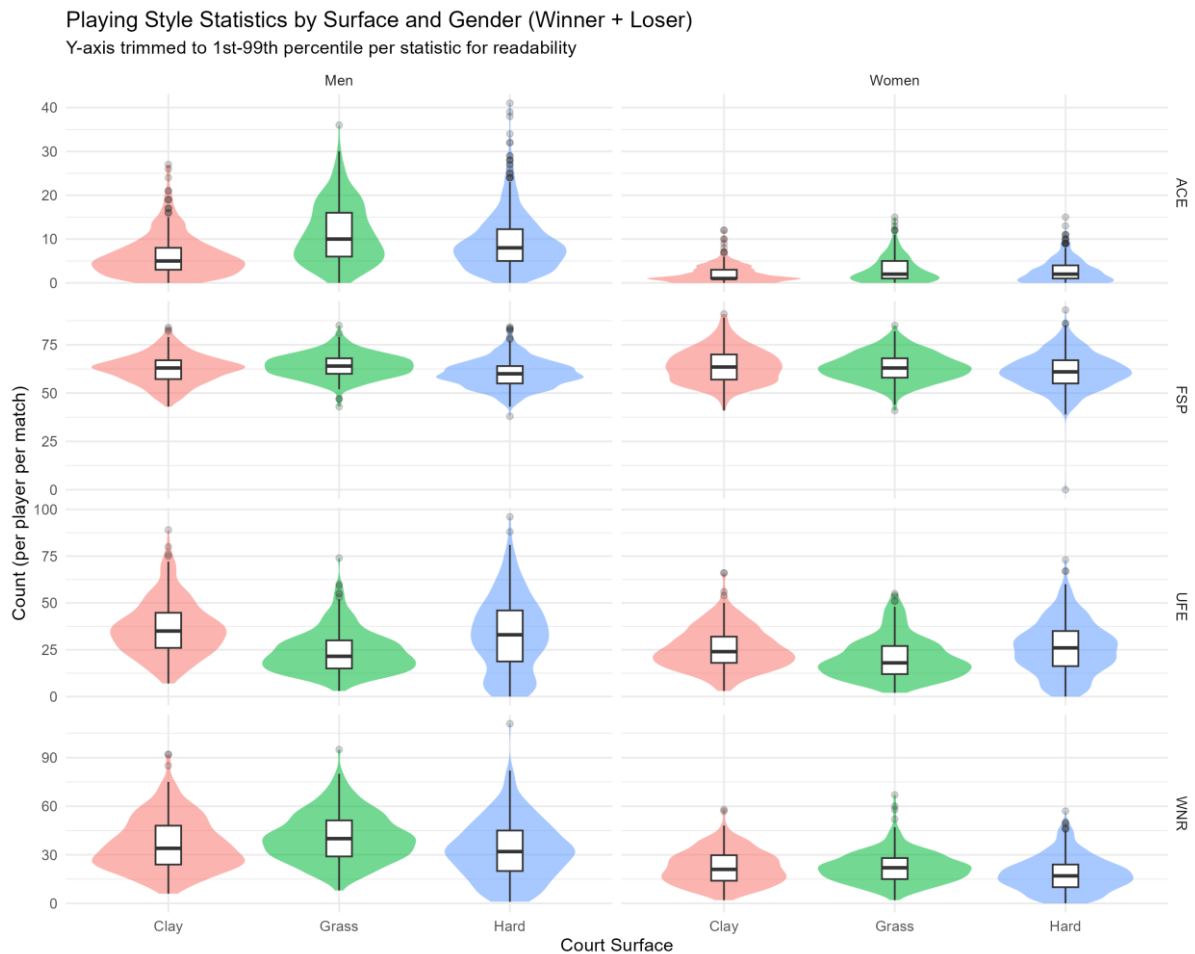


Figure 3. Distributions of ACE, FSP, UFE, and WNR by surface and gender (winner and loser combined). The y-axis is trimmed to the 1st–99th percentile per statistic for readability; this only affects the plotted range, not the underlying data.

Several surface effects are visible:

- Serving is strongest on grass, especially for men, with the highest ace counts of any surface, consistent with faster court conditions rewarding a big serve.
- Clay produces the fewest winners and the most unforced errors for men, consistent with longer, more grinding rallies on the slowest surface.
- Across all surfaces, men show substantially higher aces and winners than women, while women's matches show a smaller spread between surfaces overall.

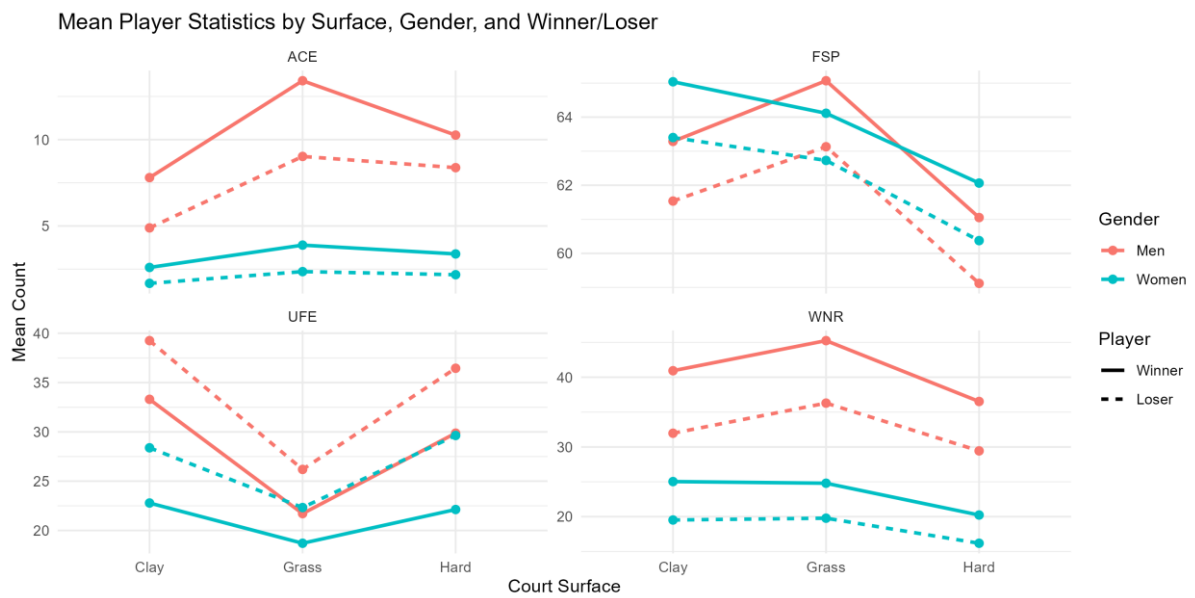


Figure 4. Mean player statistics by surface, gender, and winner/loser. This confirms the surface ordering seen in Figure 3 while adding the winner/loser split: winners are consistently ahead of the player they beat on aces and winners hit across every surface and gender, and behind them on unforced errors, which is exactly the pattern one would expect if these statistics genuinely tracked playing quality rather than just playing style.

Winning Margin: Winners vs Unforced Errors

The overall picture first, before splitting it apart by tournament:

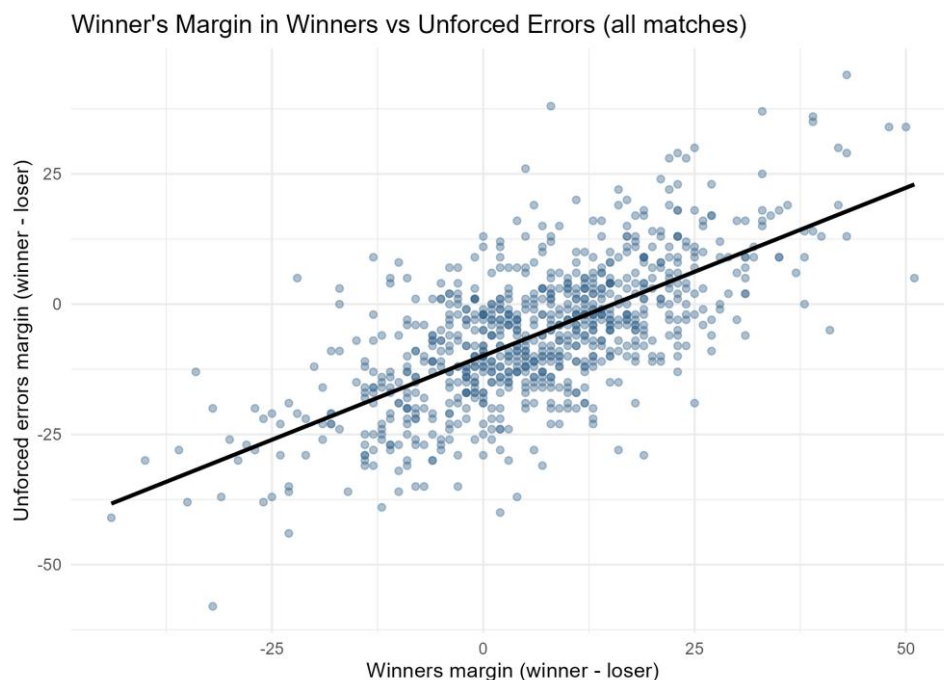


Figure 5. Winner's margin in winners hit vs unforced errors, across all matches. There is a clear positive relationship: players who beat their opponent by a wide margin in winners hit also tend to have committed noticeably fewer unforced errors than that opponent. In other words, convincing wins tend to combine both more attacking and more consistent tennis, rather than trading one off against the other.

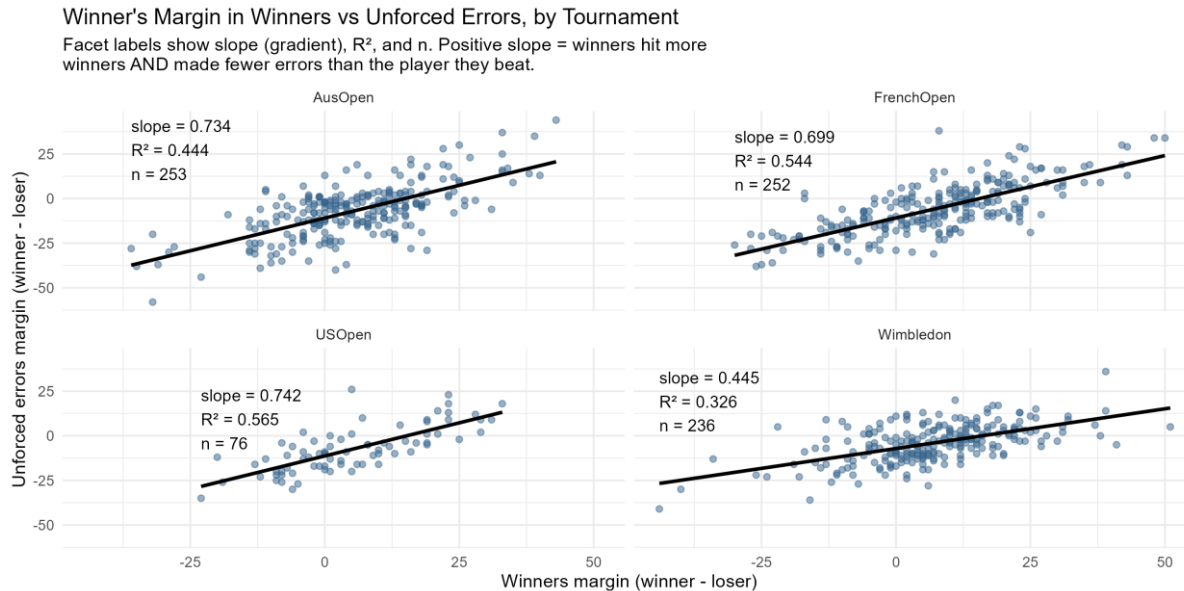


Figure 6. The same relationship, faceted by tournament, with the regression slope, R^2 , and sample size annotated per panel.

Breaking this down by tournament shows the relationship holds everywhere but with a meaningfully different strength by surface:

- The **US Open** and **Australian Open** (both Hard) show the strongest relationship (slope 0.74 and 0.73, $R^2 = 0.57$ and 0.44 respectively), so on hard courts, winning margin in errors and winners are tightly linked. The US Open's R^2 should be read cautiously given its smaller sample ($n = 76$), a direct consequence of the missingness discussed above.
- The **French Open** (Clay) is close behind (slope 0.70, $R^2 = 0.54$).

- **Wimbledon** (Grass) shows a visibly weaker relationship (slope 0.45, $R^2 = 0.33$) than any other surface. This is consistent with grass rewarding quicker, more serve-decided points, where a big winning margin doesn't have to come packaged with a correspondingly clean error count, so points can be won or lost on a handful of serve-dictated moments rather than a sustained pattern of clean, attacking play.

How the Winner's Margin Changes by Round

The US Open is excluded from this section entirely, given how sparse its round information is relative to the other three majors.

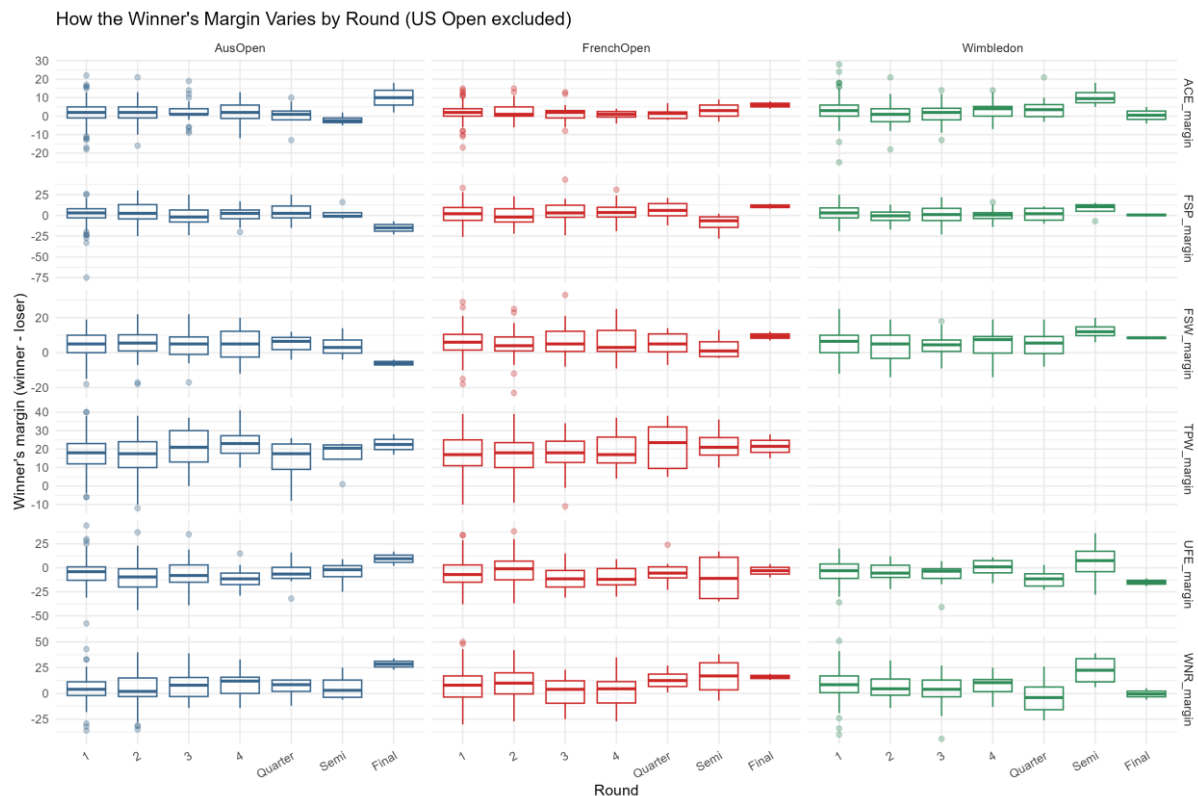


Figure 7. Winner's margin across six statistics, by round and tournament (US Open excluded).

Through the group-stage-equivalent rounds (Round 1 to the Quarter-final), margins are broadly stable to slightly narrowing, consistent with a knockout draw gradually filtering out weaker players. However, in a single elimination draw the number of matches at each stage falls mechanically: 64 matches in Round 1 down to just 2 semi-finals and 1 final per tournament, so the Semi-final and Final boxes in Figure 7 are built from very few matches and are correspondingly noisy rather than reliable estimates. The Final row for several metrics is a single match, effectively a point estimate rather than a distribution.

Standard Deviation of Winner's Margin by Round (US Open excluded)

A falling line means match outcomes become more consistent (less variable) in later rounds

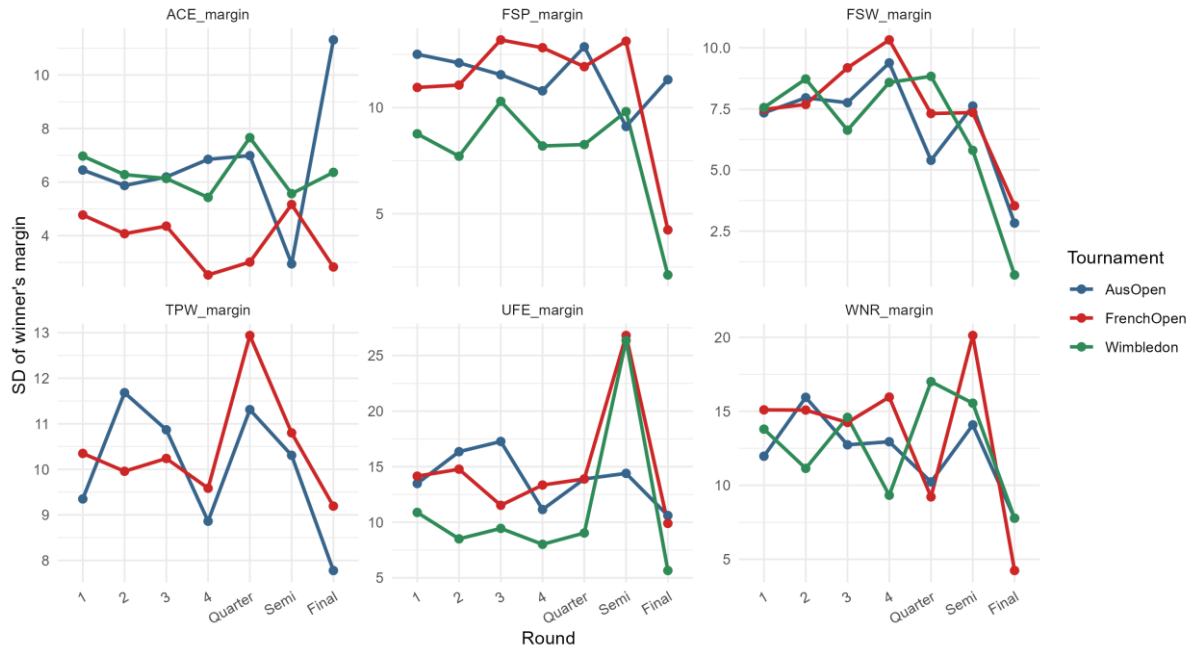


Figure 8. Standard deviation of the winner's margin by round (US Open excluded). A falling line would indicate matches becoming more consistently decided as the tournament progresses.

This plot makes the small-sample issue explicit rather than hiding it in a boxplot. The SD trend is reasonably flat and gently declining through Rounds 1–4 for most metrics, but swings sharply at the Semi-final stage for the unforced-errors margin at both the French Open and Wimbledon, which is almost certainly an artefact of computing a standard deviation from just two matches, rather than a genuine change in how matches are contested. The honest reading is that this dataset supports a modest narrowing-of-margins story through the earlier rounds but cannot support strong claims about the closing stages of each tournament.

Conclusions

- Court surface strongly shapes playing style: grass rewards aggressive, serve-dominated tennis; clay rewards consistency and rallies; hard courts sit in between but show the highest error rates.
- Winning margins in winners hit and unforced errors are positively related on every surface, but that relationship is clearly weakest on grass, supporting the idea that faster surfaces reward decisive, serve-driven points over sustained rally control.
- Match margins narrow gradually through the earlier rounds, but round-level statistics from the quarterfinals onward are built from too few matches to draw strong conclusions from, which is a limitation of the knockout format itself rather than of the data.
- Higher missingness in the US Open data, especially around round information, limits confidence in comparisons involving that tournament and is treated accordingly throughout.

Overall, the dataset supports a fairly clear surface-effects story in professional tennis, with the winner/loser recoding and the small-sample caveats above being the two adjustments that matter most for reading these figures correctly.